

Daejun Park

daejunpark@gmail.com | daejunpark.github.io | github.com/daejunpark | linkedin.com/in/daejunpark

Summary

I build practical tools and methods that make rigorous reasoning usable for real software systems.

Experience

- Andreessen Horowitz, a16z crypto** — *Senior Blockchain Security Engineer* May 2022 – Jun 2026
- Built developer-facing formal verification tools and methods, including [Halmos](#).
 - Applied formal verification to distributed protocols and financial systems.
 - Researched AI agents that identify exploitable logic flaws in decentralized finance protocols.
- Runtime Verification Inc.** — *Director of Formal Verification (previously Tech Lead)* Jan 2019 – Apr 2022
- Led formal verification and security auditing for smart-contract software and distributed protocols.
 - Identified 100+ flaws and vulnerabilities across 15+ blockchain systems.
- Microsoft Research** — *Research Intern* May 2017 – Aug 2017
- Researched SNARK-based verifiable computing for neural network training.
- Seoul National University, ROSAEC Center** — *Researcher* Jan 2012 – Jun 2013
- Researched applying homomorphic encryption to execute encrypted programs without decryption.
- Sparrow Co., Ltd.** — *Founding Engineer* Sep 2008 – Dec 2011
- Built the company's first static analysis engine for detecting memory safety bugs in embedded C programs.

Selected Contributions

Developer-Facing Formal Verification Tools

- Designed [Halmos](#) to make formal verification accessible to developers without a formal-methods background.
- [Repurposed property tests](#) as formal specifications and proved them over symbolic inputs, avoiding separate formal specs.
- Integrated seamlessly with existing developer workflows, minimizing setup friction and lowering the barrier to first use.
- Adopted by major protocol teams and used by security researchers during audits.

Formal Verification of Distributed Protocols and Financial Systems

- Covered protocols including decentralized exchanges, lending markets, consensus mechanisms, and fault-dispute systems.
- Proved both correctness and security properties, including those involving adversarial behavior and economic incentives.
- Found critical design and implementation bugs in confidential reviews, helping prevent severe failures before deployment.

AI for Automated Reasoning

- Prototyped [agentic systems](#) for automating vulnerability discovery, validating findings with proof-of-concept exploits.
- [Explored](#) how AI-assisted Lean workflows can support protocol reasoning and verification.

Executable Formal Semantics and Verification Techniques

- Built and extensively tested formal semantics for [JavaScript](#), [Ethereum Virtual Machine](#), and [x86-64](#).
- Used executable semantics as reference models for testing, interpreters, and [verification tools](#).
- Developed language-parametric [equivalence checking](#) techniques for translation validation.

Education

- University of Illinois at Urbana-Champaign**, Ph.D., Computer Science 2019
- Seoul National University**, M.S., Electrical Engineering and Computer Science 2008
- Seoul National University**, B.S., Computer Science and Engineering 2006

Awards

- Distinguished Paper Award, OOPSLA'16, ACM SIGPLAN 2016
- Feng Chen Memorial Award, University of Illinois at Urbana-Champaign 2017